Notes: Rodnyansky, Timmer, and Yago (JFE, 2026)

Intervening against the Fed

Contents

1 Big picture

- **Question.** Can foreign exchange interventions (FXIs) shield foreign economies from US monetary policy spillovers, and through which channel do they work?
- **Setting.** The paper studies FOMC announcement shocks and foreign responses in 13 countries using daily FXI data, exchange rates, country-level financial variables, and firm-level foreign stock returns.
- **Core challenge.** Both US monetary policy and FXIs are endogenous. Simply observing exchange rates or stock prices around intervention dates does not reveal the counterfactual of what would have happened without intervention.
- **Main negative fact without interventions.** A surprise US monetary tightening depreciates foreign currencies against the dollar, raises the foreign-currency risk premium, induces portfolio outflows, and lowers foreign stock prices. Firms with US dollar debt are hit much harder.
- **Main mechanism.** The paper emphasizes a **balance sheet channel**. When the local currency depreciates, firms with dollar debt face higher debt repayment in local-currency terms, which reduces net worth and lowers stock prices.
- **Main policy result.** "Intervening against the Fed"—selling US dollars after a contractionary US monetary surprise—stabilizes exchange rates, prevents the rise in currency risk premia, reduces portfolio outflows, and disproportionately cushions the stock-price decline of firms with dollar debt.
- **Key identification idea.** The paper combines high-frequency US monetary surprises with an event-study local-projection difference-in-differences design, and uses deviations from an estimated FXI reaction function to isolate the unexpected component of intervention.
- **Main empirical takeaway.** The spillover of US monetary policy is not uniform across firms. The relevant cross-sectional heterogeneity is whether firms carry US dollar debt.
- **Policy tradeoff.** FXIs help firms with dollar debt, but they can hurt firms with international sales or assets by muting the depreciation that would otherwise support foreign-currency revenues through expenditure-switching effects.
- **Why this paper matters.** The paper connects open-economy macro, international finance, and corporate balance sheets. It shows that whether a country can insulate itself from the global financial cycle depends not just on macro policy but also on firms' currency mismatch and on reserve-backed FX intervention capacity.

2 Data and measurement (key objects)

2.1 Observed country-time policy and market panel

- **Sample period.** 2000–2019.
- **Countries.** Argentina, Australia, Brazil, Chile, Colombia, Costa Rica, Georgia, Hong Kong, Japan, Mexico, Peru, Switzerland, and Turkey.

- **US monetary shocks.** The paper uses the high-frequency Fed funds rate (FFR) surprise from Nakamura and Steinsson (2018), updated by Acosta (2023), measured from changes in Fed funds futures in a 30-minute window around FOMC announcements.
- **Number of events.** The sample contains 90 FOMC announcement dates.
- **FX interventions.** Daily sterilized interventions are collected from central-bank sources, FRED, and direct requests to central banks. The analysis focuses on interventions against the US dollar.
- **Exchange rates.** Daily spot exchange rates come from Thomson Reuters Datastream. The exchange rate is defined as the value of one US dollar in local currency, so a higher value means a depreciation of the local currency.
- **Financial-channel variables.** The paper uses the VIX, a constructed currency risk premium, 10-year foreign sovereign yields, and EPFR equity fund flows to trace the risk-aversion, foreign-exchange-risk, yield-curve, and portfolio-rebalancing channels.

2.2 Observed firm-level panel

- **Stock prices.** Firm-level daily stock prices are from Datastream, with timing adjusted for whether the local market closes before or after the FOMC announcement.
- **Balance sheets and debt currency.** Corporate balance-sheet and debt-instrument data are from Capital IQ. The key advantage is that Capital IQ reports the repayment currency of each debt instrument.
- **Dollar debt measure.** The main firm-level state variable is whether firm i had US dollar debt in the previous year, denoted $\text{USD}_{i(c),y-1(t)}$. In robustness checks the paper also uses the continuous share of dollar debt in total debt.
- **Other firm controls.** One-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age, and industry-level import content of production.
- **Additional firm characteristics.** Exports, international sales, international assets, and incorporation date are from Worldscope.
- **Sample restriction.** Because stock-price data are needed, the paper focuses on publicly listed firms.

2.3 Key measured objects

- **Counter-intervention indicator.** The raw policy variable $\widetilde{FXI}_{c,t}$ takes value 1 if, after a contractionary FOMC shock, the central bank sells US dollars within the five days after the meeting; it takes value -1 for the symmetric case after a US easing shock; otherwise it is zero.
- **Unexpected intervention.** The paper estimates an FXI policy rule and uses the residual as the surprise component. The benchmark intervention dummy $FXI_{c,t}$ equals one when the absolute residual is above its median.
- **Stock-return window.** The main outcome is

$$\Delta\text{StockPrice}_{i(c),t} = y_{i(c),t+1} - y_{i(c),t-1},$$

where t is the FOMC date.

- **Exchange-rate window.** Similarly,

$$\Delta\text{ExchangeRate}_{c,t} = e_{c,t+1} - e_{c,t-1}.$$

- **Currency risk premium.** Constructed from daily exchange-rate and interbank-rate data; a higher value means investors require more compensation to hold the foreign currency against the US dollar.
- **Portfolio flows.** Weekly country-level equity flows from EPFR are used to proxy portfolio rebalancing in response to FOMC shocks.

2.4 Unobservables and timing

- **Timing of policy and firm state.** The FOMC shock occurs at date t; FXIs are measured over the following five days; firm balance-sheet heterogeneity is predetermined using the previous year's dollar-debt position.
- **Why lagged controls matter.** Macro variables such as policy rates, GDP, inflation, trade balance, and unemployment are lagged to limit simultaneity between intervention and contemporaneous macro conditions.
- **Why time zone adjustments matter.** Because local markets close at different hours, the paper shifts reported dates so that exchange-rate and stock-price observations are aligned with the FOMC announcement time.
- **Interpretation of the surprise component.** The residual from the FXI rule is meant to capture the component of intervention not forecastable from observable macro and financial conditions.

3 Models and empirical specifications (all equations + interpretation)

3.1 FXI reaction function and the identification of unexpected intervention

The paper first estimates a central-bank reaction function:

$$\widetilde{FXI}_{c,t} = \sum_{c} \beta_c \big(FFR_t \times \gamma_c\big) + \delta Z_{c,t} + \gamma_c + \varepsilon_{c,t}. \tag{1}$$

- $\widetilde{FXI}_{c,t}$ is the categorical counter-intervention indicator.
- FFR_t is the US monetary surprise on the FOMC date.
- $Z_{c,t}$ includes lagged exchange-rate trends and volatility, past interventions, lagged policy rate, GDP, inflation, trade balance, unemployment, and interactions of macro variables with the FFR shock.
- **Interpretation.** The fitted value is the expected component of intervention; the residual $\varepsilon_{c,t}$ is the unexpected component.

- **Key result from this step.** Only about 24% of the variation in counter-intervention is explained by observables, so most intervention variation is left unexplained and can be treated as surprise-like.

3.2 Baseline local-projection event study

The core event-study regression is

$$y_{i(c),t+h} - y_{i(c),t-1} = \beta^h FFR_t + X\delta_x^h + \alpha_{i(c)}^h + \varepsilon_{i(c),t}^h, \qquad h \in [-5, 5]. \tag{2}$$

- The dependent variable is the change in log stock price from the day before the FOMC meeting to horizon h after the meeting.
- The local-projection structure traces the full path of the response for $h = -5, \ldots, 5$.
- **Interpretation.** $\beta^h < 0$ means that a contractionary US monetary policy shock lowers foreign stock prices at horizon h.
- **Why the lead terms matter.** Estimates for $h < 0$ serve as a pre-trend diagnostic for the parallel-trends assumption in the LP-DID setup.

3.3 Balance sheet channel from firm-level dollar debt

To test whether firms with dollar debt are differentially exposed, the paper estimates

$$y_{i(c),t+h} - y_{i(c),t-1} = \gamma^h FFR_t \times USD_{i(c),y-1(t)} + X\delta_x^h + \alpha_{i(c)}^h + \alpha_{c,t}^h + \varepsilon_{i(c),t}^h, \qquad h \in [-5, 5]. \tag{3}$$

- The country-time fixed effect $\alpha_{c,t}^h$ absorbs all country-level shocks around the FOMC date.
- **Interpretation.** $\gamma^h < 0$ means that firms with dollar debt underperform firms without dollar debt after a US tightening shock.
- **Economic meaning.** This is the balance-sheet channel: depreciation raises the local-currency value of dollar liabilities and depresses equity values of exposed firms.

3.4 FXIs in the difference-in-differences specification

To test whether intervention mitigates the spillover, the paper estimates

$$y_{i(c),t+h} - y_{i(c),t-1} = \beta^h FFR_t + \Omega^h FXI_{c,t} + \gamma^h FFR_t \times FXI_{c,t} + X\delta_x^h + \alpha_{i(c)}^h + \varepsilon_{i(c),t}^h, \qquad h \in [-5, 5]. \tag{4}$$

- **Interpretation.** $\gamma^h > 0$ means intervention reduces the stock-price decline caused by a contractionary US monetary shock.
- **Why this is LP-DID.** The paper compares treated and untreated observations over horizons before and after each FOMC event, rather than relying on a single two-way-fixed-effect estimator.

3.5 Triple interaction: FXIs and dollar-debt firms

The most informative specification combines the two dimensions of heterogeneity:

$$\begin{aligned} y_{i(c),t+h} - y_{i(c),t-1} = \theta^h FFR_t \times USD_{i(c),y-1(t)} \times FXI_{c,t} + \gamma^h FFR_t \times USD_{i(c),y-1(t)} \\ + X\delta_x^h + \alpha_{i(c)}^h + \alpha_{c,t}^h + \varepsilon_{i(c),t}^h, \qquad h \in [-5, 5]. \end{aligned} \tag{5}$$

- γ^h captures the balance-sheet spillover when the country does *not* counter-intervene.
- θ^h captures the additional effect of intervention for firms with dollar debt.
- **Interpretation.** If $\theta^h + \gamma^h = 0$, intervention fully mutes the balance-sheet channel.

3.6 Financial-channel controls

To study whether the balance-sheet result is really just proxying for other financial channels, the paper augments the stock-price regression with country-level controls such as the VIX, the currency risk premium, foreign bond yields, and equity flows:

$$y_{i(c),t+1} - y_{i(c),t-1} = \beta FFR_t + \Omega FXI_{c,t} + \gamma FFR_t \times FXI_{c,t} + X\delta_x + D\delta_d + \alpha_{i(c)} + \varepsilon_{i(c),t}. \tag{6}$$

- **Purpose.** This asks whether the dollar-debt result survives even after controlling for other transmission channels of US monetary policy.
- **Main message.** Those controls explain a good share of the effect on firms without dollar debt, but the differential response of firms with dollar debt remains strong.

3.7 Exchange-rate regression

For exchange rates, the dependent variable is the change in the log exchange rate and the paper reuses the event-study logic of (2) and (4). The pooled baseline is

$$\Delta e_{c,t} = \beta FFR_t + \Omega FXI_{c,t} + \gamma FFR_t \times FXI_{c,t} + W\lambda + u_{c,t}, \tag{7}$$

where $\Delta e_{c,t}$ is the two-day change in the exchange rate and W includes lagged exchange-rate behavior, past interventions, and lagged macro controls.

- **Interpretation.** $\beta > 0$ means that US tightening depreciates the local currency; $\gamma < 0$ means intervention offsets that depreciation.

3.8 Other transmission channels at the country level

The paper also re-estimates the event-study equations with four country-level dependent variables:

- the **VIX**, as a measure of global risk aversion;
- the **currency risk premium**, as the compensation required to hold foreign currency risk;
- the **foreign 10-year bond yield**, as a discount-rate channel;
- **equity capital outflows**, as a portfolio-rebalancing channel.

- **Interpretation.** If FXIs work mainly through exchange-rate stabilization, they should mute the currency-risk-premium and portfolio-flow channels, but not necessarily the VIX itself.

3.9 Exporters, international revenues, and assets

The paper studies whether FXIs can hurt firms that benefit from depreciation. It re-estimates the stock-price regressions for exporters and non-exporters and also excludes firms with exports, international sales, and international assets.

- **Interpretation.** If FXIs stabilize the exchange rate, they can remove the expenditure-switching benefit from depreciation while leaving the negative demand effect from US tightening in place.

3.10 Continuous intervention size

Instead of the categorical intervention dummy, the paper also uses standardized FXI volume measured as a ratio to GDP.

- **Purpose.** This maps the economic size of intervention needed to offset the stock-market spillover to dollar-debt firms.

3.11 Reserve heterogeneity

Finally, the paper splits the sample by whether FX reserves are above or below the median.

- **Interpretation.** Larger reserve buffers make intervention more credible and more effective because the market expects the central bank to be able to continue defending the currency if needed.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- **FXIs are endogenous.** Central banks may intervene exactly when markets are stressed, so raw intervention dates are contaminated by omitted news and reverse causality.
- **Quarterly or monthly aggregation is problematic.** High-frequency US monetary shocks are small, and aggregating them to low frequency introduces serial correlation and weakens power.
- **Country-level comparisons alone are too coarse.** Cross-country stock-index responses cannot cleanly isolate firm-level balance-sheet heterogeneity.

4.2 Why the high-frequency US monetary shock helps

- The Fed funds futures surprise is measured in a narrow 30-minute window around the FOMC announcement.

- This makes it much more plausible that the shock is orthogonal to slow-moving macro conditions in the foreign country.
- Because stock prices and exchange rates are forward looking, daily responses around the event provide a clean read on the immediate transmission of US policy.

4.3 Why the FXI policy-rule residual helps

- The policy-rule residual strips out the predictable component of intervention driven by past exchange-rate movements, past interventions, and lagged macro conditions.
- The low R^2 of the FXI rule is helpful for identification: most intervention variation remains unexplained by observables.
- The benchmark definition of unexpected intervention focuses on large residuals in absolute value, which is meant to isolate episodes in which the central bank moved more than the market would have expected.

4.4 Why firm-level dollar debt is the key source of identification

- The firm-level interaction with dollar debt allows the paper to ask a cross-sectional question within the same country and date: do dollar-debt firms react more negatively to the same US monetary shock?
- Once country-date fixed effects are included, the remaining identification comes from within-country variation across firms.
- That makes the balance-sheet interpretation much more credible than a pure country-level regression.

4.5 Why the LP-DID design is informative

- Estimating separate horizons before and after the shock allows the paper to inspect pre-trends directly.
- Near-zero coefficients for $h < 0$ support the parallel-trends assumption.
- Staggered-treatment bias is less of a concern because the event window is only 10 days, while FOMC meetings occur roughly every six weeks, so treated units are not reused as controls in overlapping windows.

4.6 Why the other channel controls matter

- The paper explicitly controls for the VIX, the currency risk premium, foreign yields, and equity flows.
- If the dollar-debt interaction survives those controls, then the mechanism is not just a relabeling of broad global risk sentiment or portfolio rebalancing.

4.7 Remaining limitations

- The FXI-rule residual is not a perfect instrument. Unobserved investor sentiment or news can still affect both intervention and asset prices.
- The analysis covers only publicly listed firms, so the paper does not observe how real outcomes of private firms respond.
- Firm-level data on invoicing currency, derivative hedging, and detailed destination exposure are limited, so the expenditure-switching channel is harder to test cleanly than the dollar-debt channel.
- Intraday intervention timing is unavailable for most countries, preventing a cleaner identification of the within-day causal effect of intervention.

5 Tables and figures

5.1 Figure 1: Bank of Japan example

Read:

- The figure shows a concrete episode on September 22, 2022: after a 75 bp FOMC tightening, the yen depreciated, and then appreciated sharply once the Bank of Japan sold US dollars.
- This is not the paper's identification design by itself, but it gives intuition for the phrase "intervening against the Fed."
- It visually motivates the idea that FXI can offset US monetary spillovers in the exchange rate.

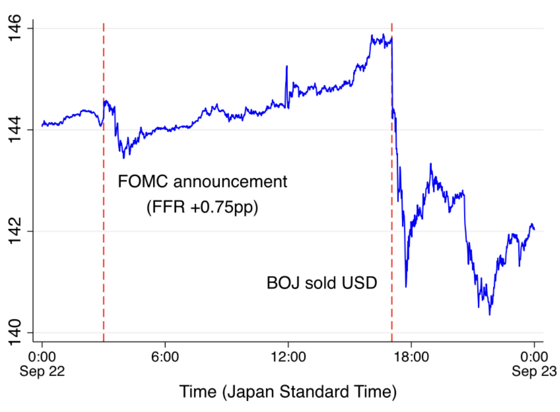

Fig. 1. Spot exchange rate: US Dollar to Japanese Yen.
Note: The figure reports the minute-by-minute US Dollar to Japanese yen spot exchange rate on September 22, 2022. The exchange rate is defined as the value of one US Dollar in terms of yen, and a higher value implies the appreciation of the Dollar or depreciation of Japanese yen.
Source: Datastream.

Table 1
Stock price: Baseline regression.

(a) Without intervention				
Dependent variable:	ΔStock Price$_{i(c),t}$			
	Dollar debt	No dollar debt	Both	
	(1)	(2)	(3)	(4)
FFR Shock$_t$	−0.660***	−0.094**	−0.097**	
	(0.117)	(0.045)	(0.045)	
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$			−0.314***	−0.259***
			(0.087)	(0.071)
R^2	0.093	0.032	0.031	0.083
N	1926	103,155	105,114	105,114
Firm FE	✓	✓	✓	✓
Country × Date FE				✓

(b) With intervention				
Dependent variable:	ΔStock Price$_{i(c),t}$			
	Dollar debt	No dollar debt	Both	
	(1)	(2)	(3)	(4)
FFR Shock$_t$	−0.217**	−0.149***	−0.158**	
	(0.105)	(0.056)	(0.061)	
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$			−0.001	−0.033
			(0.042)	(0.035)
R^2	0.114	0.209	0.194	0.270
N	1258	9915	11,178	11,178
Firm FE	✓	✓	✓	✓
Country × Date FE				✓

Note: ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to $t+1$, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. Dollar Debt$_{i(c),y-1(t)}$ is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

5.2 Table 1, Figure 2, and Figure 3: The balance-sheet channel in stock prices

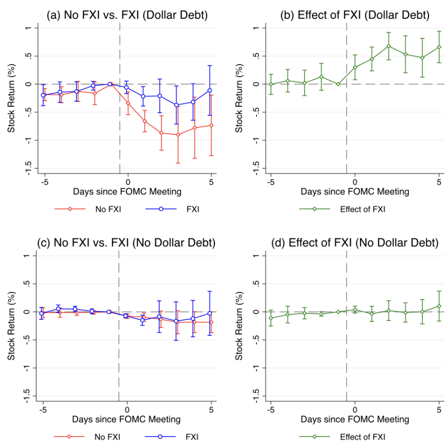

Fig. 2. Stock price: Difference-in-difference.
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices. See Eqs. (2) and (4) for the exact specifications. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are double clustered at the firm and date level. The confidence interval is 90%. The red and blue lines show the results for countries with and without FXIs, respectively, and the green line shows the difference between them.

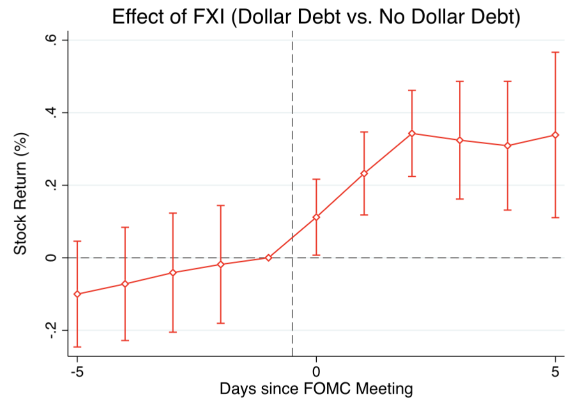

Fig. 3. Stock price: Triple interaction.
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices. See Eq. (5) for the exact specifications. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The confidence interval is 90%.

Read:

- Without intervention, a 10 bp surprise increase in the Fed funds rate lowers the stock price of firms with dollar debt by about 6.6%, while the effect for firms without dollar debt is only about 0.9%. The interaction term is about -0.314 in the pooled regression and about -0.259 even with country-date fixed effects.
- With intervention, the stock-price decline of firms with dollar debt is much smaller—about 2.2%—and the differential effect relative to firms without dollar debt disappears statistically.
- Figure 2 shows the dynamic path: without FXI the stock-price decline accumulates to almost 10% after about three days, while with FXI the effect is much smaller and largely disappears by day five.
- Figure 3 plots the triple interaction and shows that the mitigating effect of FXI is systematically stronger for firms with dollar debt. In the text's interpretation, the triple interaction roughly offsets the dollar-debt coefficient, implying that FXIs nearly fully mute the balance-sheet channel.

5.3 Table 2: Other financial channels without intervention

Read:

- For firms without dollar debt, the effect of US tightening is modest and becomes insignificant once all four financial-channel controls are included simultaneously.
- For firms with dollar debt, the effect remains large and strongly significant even after controlling for the VIX, currency risk premium, foreign bond yield, and capital flows. The estimated coefficient stays in the range of roughly -0.55 to -0.66.
- This is strong evidence that the main differential effect for dollar-debt firms is not just broad risk sentiment or portfolio rebalancing.

Table 2
US monetary spillover without interventions (Controlling for financial channels).

(a) Firms without US dollar debt

Dependent variable	ΔStock Price$_{i(c),t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock$_t$	−0.094**	−0.030	−0.036	−0.093**	−0.094**	0.009
	(0.045)	(0.034)	(0.044)	(0.045)	(0.045)	(0.037)
R^2	0.032	0.044	0.042	0.035	0.032	0.054
N	103,155	103,155	95,668	103,155	103,155	95,668
VIX		✓				✓
Currency risk premium			✓			✓
Foreign yield				✓		✓
Capital flows					✓	✓
Firm FE	✓	✓	✓	✓	✓	✓

(b) Firms with US dollar debt

Dependent variable	ΔStock Price$_{i(c),t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock$_t$	−0.660***	−0.626***	−0.580***	−0.666***	−0.549***	−0.552***
	(0.117)	(0.109)	(0.117)	(0.111)	(0.126)	(0.130)
R^2	0.093	0.104	0.113	0.094	0.100	0.127
N	1926	1926	1724	1926	1926	1724
VIX		✓				✓
Currency risk premium			✓			✓
Foreign yield				✓		✓
Capital flows					✓	✓
Firm FE	✓	✓	✓	✓	✓	✓

Note: ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to $t+1$, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. In column (1), we do not control for the VIX index, currency risk premium, government bond yield, and equity flows. In column (2), we control for the VIX index. In column (3), we control for the currency risk premium. In column (4), we control for the government bond yield. In column (5), we control for equity flows. In column (6), we control for the VIX index, currency risk premium, government bond yield, and equity flows at the same time. We take the sample firms in countries without counter-interventions. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double-clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

- In other words, the balance-sheet channel is not subsumed by the paper's other financial channels.

5.4 Table 3 and Figure 4: Risk premium and portfolio-rebalancing channels

Table 3
Balance-sheet channel: Firms with dollar debt (Controlling for financial channels).

Dependent variable	ΔStock Price$_{i(c),t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock$_t$	−0.647***	−0.561***	−0.600***	−0.652***	−0.558***	−0.487***
	(0.116)	(0.100)	(0.120)	(0.112)	(0.131)	(0.127)
FFR Shock$_t$ × Intervention$_{c,t}$	0.449***	0.386***	0.415***	0.452***	0.360**	0.330**
	(0.130)	(0.144)	(0.132)	(0.129)	(0.142)	(0.152)
R^2	0.091	0.100	0.111	0.092	0.097	0.120
N	3206	3206	2635	3206	3206	2635
VIX		✓				✓
Currency risk premium			✓			✓
Foreign yield				✓		✓
Capital flows					✓	✓
Firm FE	✓	✓	✓	✓	✓	✓

Note: ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to $t+1$, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. Intervention$_{c,t}$ is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. In column (1), we do not control for the VIX index, currency risk premium, government bond yield, and equity flows. In column (2), we control for the VIX index. In column (3), we control for the currency risk premium. In column (4), we control for the government bond yield. In column (5), we control for equity flows. In column (6), we control for the VIX index, currency risk premium, government bond yield, and equity flows at the same time. We take the sample firms with dollar debt. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

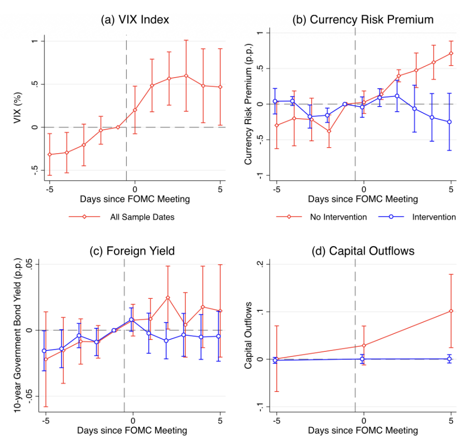

Fig. 4. Different transmission channels of FXI.
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on the VIX index, currency risk premium, foreign bond yield, and equity capital flows. In panel (a), the dependent variable is the change in the 3-month US VIX index. In panel (b), the dependent variable is the change in currency risk premium. A positive value implies a higher excess return on the foreign currency over the US dollar. In panel (c), the dependent variable is the change in foreign 10-year government bond yields. In panel (d), the dependent variable is the equity capital flows. A positive value implies a capital outflow from foreign countries. We control for the standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Standard errors are double clustered at the country and date level (panel b) and clustered at the date level (panels a, c, and d). Note that, given low clusters due to data availability, we cannot double cluster in panels (c) and (d). The confidence interval is 90%.

Read:

- Figure 4 shows that a contractionary US monetary shock raises the VIX, raises the foreign-currency risk premium, and induces equity outflows from foreign countries.
- FXIs do *not* meaningfully affect the VIX, which makes sense because the VIX is a US/global risk-aversion object rather than a country-specific exchange-rate object.
- FXIs *do* mute the rise in the currency risk premium and the portfolio outflows, which is consistent with intervention stabilizing exchange-rate risk.

- The foreign-yield channel is relatively muted. Table 3 then shows that even after controlling for these other channels, the stock-price cushioning effect of FXI for dollar-debt firms remains statistically significant: the interaction term stays around 0.33 to 0.45.

5.5 Table 4 and Figure 5: Exchange-rate stabilization

Table 4
Exchange rate: Baseline regression.

Dependent variable:	ΔExchange Rate$_{c,t}$		
	No intervention (1)	Intervention (2)	Both (3)
FFR Shock$_t$	0.225***	0.004	0.201**
	(0.069)	(0.021)	(0.072)
Intervention$_{c,t}$			0.266
			(0.155)
FFR Shock$_t$ × Intervention$_{c,t}$			−0.202**
			(0.072)
R^2	0.108	0.083	0.084
N	418	417	836
Country FE	✓	✓	✓

Note: ΔExchange Rate$_{c,t}$ is the change in the log of the exchange rate from date $t-1$ to $t+1$, where t is the FOMC announcement date. The exchange rate is defined as the value of the US dollar in terms of local currency, and a higher value implies a depreciation of the local currency. FFR Shock$_t$ is the Fed funds rate shock in basis points. Intervention$_{c,t}$ is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. We control for the trend and standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, the one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the country and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

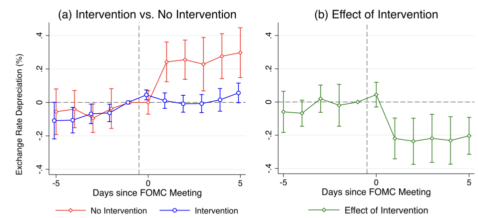

Fig. 5. Exchange rate: Difference-in-differences.
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on exchange rates. See Eq. (4) for the exact specification. The exchange rate is defined as the value of the US dollar in terms of local currency, and a higher value implies a depreciation of the local currency. We control for the standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Standard errors are double clustered at the country and date level. The confidence interval is 90%.

Read:

- Without intervention, a 10 bp surprise increase in the Fed funds rate depreciates the local currency by about 2.25%.
- With intervention, the same depreciation is essentially zero. In the pooled regression, the interaction term is about −0.202, meaning FXIs offset roughly 2.0 percentage points of depreciation relative to the no-intervention case.
- Figure 5 shows the dynamic counterpart: countries without FXI exhibit a persistent post-FOMC depreciation, while countries with FXI remain roughly flat.
- This is the macro counterpart of the firm-level balance-sheet result: FXIs help dollar-debt firms precisely because they prevent the exchange-rate depreciation that would otherwise inflate local-currency debt burdens.

5.6 Table 5 and Figure 6: The expenditure-switching channel and the policy tradeoff

Read:

- Exporters may weakly benefit from the depreciation caused by US tightening, because foreign-currency revenues become more valuable in local currency. But the estimated positive effect without FXI is small and not very precise.
- When the country intervenes and prevents depreciation, exporters can lose that expenditure-switching benefit while still suffering from weaker foreign demand.

Table 5
Stock price (Excluding firms with foreign revenues and assets).

(a) Without intervention

Dependent variable	ΔStock Price$_{i(c),t}$				
	(1)	(2)	(3)	(4)	(5)
FFR Shock$_t$	−0.097**	−0.095**	−0.089**	−0.091*	−0.082*
	(0.045)	(0.044)	(0.044)	(0.046)	(0.043)
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$	−0.314***	−0.316***	−0.292***	−0.313***	−0.313***
	(0.087)	(0.088)	(0.082)	(0.088)	(0.085)
R^2	0.031	0.032	0.035	0.035	0.036
N	105,114	96,056	93,608	97,478	83,128
Exclude firms with exports		✓			✓
Exclude firms with international sales			✓		✓
Exclude firms with international assets				✓	✓
Firm FE	✓	✓	✓	✓	✓

(b) With intervention

Dependent variable	ΔStock Price$_{i(c),t}$				
	(1)	(2)	(3)	(4)	(5)
FFR Shock$_t$	−0.158**	−0.157**	−0.089	−0.072	−0.085
	(0.061)	(0.062)	(0.082)	(0.049)	(0.077)
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$	−0.001	0.002	−0.044	−0.015	−0.041
	(0.042)	(0.045)	(0.081)	(0.059)	(0.086)
R^2	0.194	0.186	0.257	0.205	0.246
N	11,178	10,640	6153	9713	5606
Exclude firms with exports		✓			✓
Exclude firms with international sales			✓		✓
Exclude Firms with International Assets				✓	✓
Firm FE	✓	✓	✓	✓	✓

Note: ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to $t+1$, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. Dollar Debt$_{i(c),y-1(t)}$ is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. In column (2), we exclude firms with exports. In column (3), we exclude firms with international sales. In column (4), we exclude firms with international assets. In column (5), we exclude firms with exports, international sales, and international assets. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

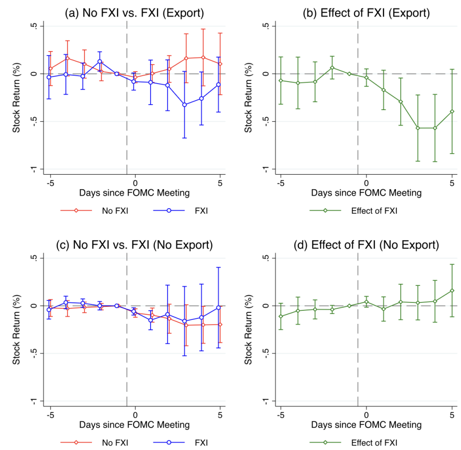

Fig. 6. Stock price: Expenditure switching channel.
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices for exporters and non-exporters. See Eqs. (3) and (5) for the exact specifications. We control for one-year lagged dollar debt, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are double clustered at the firm and date level. The confidence interval is 90%.

- The negative effect of FXI on firms without dollar debt becomes statistically insignificant once the sample excludes firms with international sales or international assets. That suggests that the apparent downside of intervention for non-dollar-debt firms is concentrated in firms with foreign revenues or foreign assets.
- The paper's policy message is therefore not one-sided: FXIs help firms with dollar debt, but can hurt firms whose balance sheets or revenue models benefit from a weaker local currency.

5.7 Table 6: How much intervention is needed?

Read:

- Replacing the intervention dummy with standardized FXI volume yields a similar qualitative result: larger US dollar sales by the central bank offset the stock-price decline of dollar-debt firms.
- In the paper's calibration, a one-standard-deviation intervention volume corresponds to roughly $90 million when evaluated at the sample's median GDP, and this offsets about 1.4 percentage points of the relative stock-price decline.
- The paper notes that roughly $101 million of dollar sales would fully offset the stock-market spillover through the balance-sheet channel.
- That is economically meaningful because observed country-specific intervention volumes are of comparable order of magnitude in the sample.

5.8 Table 7: FX reserves and the effectiveness of FXIs

Read:

- FXIs work better when countries have larger reserve buffers.

Table 6
Stock price: FXI volume.

Dependent variable	ΔStock Price$_{i(c),t}$	
	(1)	(2)
FFR Shock$_t$	−0.097**	
	(0.045)	
FFR Shock$_t$ × Intervention$_{c,t}$	0.005	
	(0.063)	
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$	−0.152***	−0.167***
	(0.044)	(0.046)
FFR Shock$_t$ × Intervention$_{c,t}$ × Dollar Debt$_{i(c),y-1(t)}$	0.142**	0.111**
	(0.058)	(0.051)
R^2	0.034	0.086
N	114,448	114,448
Firm FE	✓	✓
Country × Date FE		✓

Note: ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to $t+1$, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. Intervention$_{c,t}$ is the standardized volume of unexpected intervention. The FXI volume is measured in terms of the percentage ratio over GDP. A positive value indicates a net purchase of the local currency. Dollar Debt$_{i(c),y-1(t)}$ is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double-clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table 7
FX reserves.

Dependent variable:	ΔExchange Rate$_{c,t}$		ΔStock Price$_{i(c),t}$			
FX reserve:	Large	Small	Large	Small	Large	Small
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock$_t$	0.252***	0.122	−0.075	−0.126		
	(0.065)	(0.071)	(0.067)	(0.094)		
FFR Shock$_t$ × Intervention$_{c,t}$	−0.298***	−0.161*	−0.108	0.162		
	(0.089)	(0.080)	(0.079)	(0.143)		
FFR Shock$_t$ × Dollar Debt$_{i(c),y-1(t)}$			−0.387**	−0.261***	−0.352***	−0.230***
			(0.151)	(0.086)	(0.108)	(0.079)
FFR Shock$_t$ × Intervention$_{c,t}$ × Dollar Debt$_{i(c),y-1(t)}$			0.418**	0.137	0.317***	0.201*
			(0.172)	(0.112)	(0.118)	(0.102)
R^2	0.121	0.149	0.053	0.051	0.109	0.098
N	422	413	90,860	25,880	90,860	25,880
Country FE	✓	✓				
Firm FE			✓	✓	✓	✓
Country × Date FE					✓	✓

Note: Columns (1) and (2) show the results for country-level exchange rates, and columns (3) to (6) show the results for firm-level stock prices. ΔExchange Rate$_{c,t}$ is the change in the log of the exchange rate from date $t-1$ to $t+1$, and ΔStock Price$_{i(c),t}$ is the change in the log of firm-level stock prices from date $t-1$ to $t+1$, where t is the FOMC announcement date. FFR Shock$_t$ is the Fed funds rate shock in basis points. Intervention$_{c,t}$ is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. Dollar Debt$_{i(c),y-1(t)}$ is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. Large and small FX reserves are defined so that the volume of FX reserves is larger and smaller than the median, respectively. In columns (1) and (2), we control for the trend and standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. In columns (3)–(6), we control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

- In countries with large FX reserves, a 10 bp Fed hike leads to about a 2.5% exchange-rate depreciation without intervention, but the depreciation is fully stabilized when the central bank intervenes.
- For stock prices, the mitigating effect of FXI on the balance-sheet channel is much larger in high-reserve countries. The triple interaction is about 0.418 without country-date fixed effects and 0.317 with them, versus much smaller values in low-reserve countries.
- This supports the idea that accumulated reserves provide a self-insurance mechanism that makes intervention more credible and more potent.

5.9 Appendix robustness

Read:

- The paper shows robustness to alternative definitions of unexpected intervention, continuous intervention size, sterilized interventions, debt maturity, intensive and extensive margins of dollar debt, and excluding the global financial crisis and zero-lower-bound episodes.
- The balance-sheet result is also robust to controls for hedging, international sales and assets, and a variety of cross-country characteristics such as development, trade openness, and capital-account openness.

6 Short summary

- **Conclusion.** A surprise US monetary tightening spills over to foreign stock prices mainly by depreciating foreign currencies and hurting firms with US dollar debt. Counter-intervening in the FX market can largely neutralize that channel.

- **Identification insight.** The paper combines high-frequency US monetary surprises, an FXI policy-rule residual, firm-level dollar-debt heterogeneity, and an LP-DID design to isolate the causal effect of "intervening against the Fed."
- **Empirical lesson.** FXIs stabilize exchange rates, currency risk premia, and portfolio flows, and this is especially valuable for firms with dollar debt. But the same policy can work against exporters and firms with international revenues by limiting the gains from depreciation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that US monetary spillovers propagate to foreign equity prices through corporate currency mismatch.
- It also establishes that FXIs can be an effective insulating tool: by selling US dollars after a contractionary FOMC surprise, central banks can stabilize their currencies and thereby cushion the stock-price decline of firms with dollar debt.
- This result survives controls for other financial channels, alternative measures of intervention, and a wide range of robustness exercises.

7.2 Economic intuition

- A surprise US tightening strengthens the dollar. For a firm that owes debt in dollars but earns revenues in local currency, this is bad news: the local-currency value of liabilities jumps, net worth falls, and equity prices fall.
- If the central bank intervenes and prevents the depreciation, it effectively insures those firms against the currency mismatch on their balance sheets.
- But exchange-rate stabilization is not free. The same depreciation that hurts dollar-debt firms can help exporters and firms with foreign revenues. That is why FXI creates a tradeoff across firms.
- Reserve accumulation matters because intervention only works well if markets believe the central bank has enough firepower to continue stabilizing the currency.

7.3 Takeaway

This paper is best viewed as a high-frequency micro-founded study of how countries can resist the global financial cycle. Its central message is that FXIs are not just about smoothing the exchange rate mechanically. They matter because they reshape the transmission of US monetary policy to corporate balance sheets, foreign-currency risk premia, and international portfolio flows. Once firm-level currency mismatch is brought to the center of the analysis, the effectiveness of "intervening against the Fed" becomes both economically intuitive and empirically visible.